

A
WINTER PROJECT REPORT
ON
“A STUDY ON IMPACT OF INDIVIDUAL EQUITY INVESTOR’S BEHAVIOURAL
BIASES TO INVESTMENT DECISION ON EQUITY MARKET IN SURAT REGION”

In partial fulfillment of the requirement of the award for the degree of

Bachelor of Business Administration

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Submitted to:



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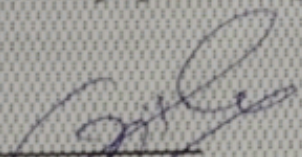
CERTIFICATE

This is to certify that the project entitled "A Study on Impact
of individual equity investors' Behaviour in
Investment Decision on equity market"
has been carried out by Mr./Ms. Vaid Urmila

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towards fulfillment of the course, for the Academic Year 2023-24.


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ACKNOWLEDGEMENT

I am highly obligated to the VEER NERMAD SOUTH GUJARAT UNIVERSITY and management of SHRI SHAMBHIBHAI V. PATEL COLLEGE OF COMPUTER SCIENCE AND BUSINESS MANAGEMENT SURAT firstly, I wish my heartfelt thanks to In-charge principal of this college **Mr. Chintu Thakkar**, who always motivate us to fulfill us dreams and lead us in the direction of growth. He has given his valuable inputs for effective completion of this project. And special thanks of my guide, **PROF. SWATI HALANI** for their efforts, guidance and give her valuable time to suggestion for making project of “**A STUDY ON IMPACT OF INDIVIDUAL EQUITY INVESTOR’S BEHAVIOURAL BIASES TI INVESMENT DECISION ON EQUITY MARKET IN SURAT REGION**” who also help me in completing my winter project. I come to know about so many new things I am thankful them.

I hereby thank all the respondents for providing me with valuable information for caring out this project.

[VALA URMILA]

TYBBA

SEM 6

DECLARATION

I am VALA URMILA hereby declare that the following project report titled “**A STUDY ON IMPACT OF INDIVIDUAL EQUITY INVESTOR’S BEHAVIOURAL BIASES TI INVESMENT DECISION ON EQUITY MARKET IN SURAT REGION**” under the guidance of **PROF. SWATI HALANI**, submitted in partial fulfillment of the requirement of the award for the marks of Bachelor of Business Administration to Veer Narmad South Gujarat University, Surat is my original work-research-study carried out during sixth semester.

PLACE: Surat

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EXECUTIVE SUMMARY

The purpose of this study was exploring impact of equity investor's behavioural biases in investment decision on equity market in Surat region. Individual investors are small time investors who invest in the stock market to create financial wealth. They usually invest the savings from their income in the equity market. An attempt has also been made to make a demographic profile of investors affected by various behavioural biases.

This report includes five chapters:

The first part is introduction of report; it includes history, all information about equity market, and also include introduction and overview of Risk and Return, financial investment and it delineates various behaviour biases like, loss aversion, overconfidence bias, regret aversion bias, representative bias, anchoring bias, confirmation bias, herd behaviour.

The literature review delves into various studies conducted on the impact of equity investor's behavioural biases to investment decision on equity market across different cities.

The research methodology outlines a systematic approach of studying the preferences of individual equity investors. It utilizes a prospective observational design with a sample size of 150 respondents from Surat city, India. Data analysis tools and techniques are mentioned and limitation of study.

The next part of this research is analysis of data, in research has used analytical tools and techniques like frequency statistics, descriptive statistics, t- test and ANOVA analysis.

The finding indicates the find out some important facts which are explained in finding after survey. Conclusion is part of project report; conclusion highlights the study's impact on investment decision and behavioural biases based on equity investment.

The last part of the report covers bibliography and annexure; BIBLIOGRAPHY includes the details of the sources of the information which help to generate this report and ANNEXURE contain the questionnaire used in survey aimed behavioural biases impact of investment decision of equity investors.

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CHAPTER:-1

Introduction

1.1 INTRODUCTION OF EQUITY MARKET:

1.1.1 History of Equity Markets:

Equity markets come with long entrenched histories, with debt issuances dating back to the 1300s. That first stock market was established in Belgium in 1531. The exchange dealt primarily with promissory notes and bonds, but not with actual stocks.

Throughout the 1600s, the British, Dutch and French governments gave charters to companies that included 'East India' in their monikers. The countries would take stakes in the profits from India and Asia by funding sea voyages that would bring back goods- although it was risky due to the abundance of pirates, poor weather and faulty navigation.

Instead of bearing all the risk for themselves, ship owners would seek out investors to help fund voyages and in return, provide investors with a percentage of the profits should the voyages be successful.

They were the earliest forms of limited liability companies (LLCs) that would last a single voyage. Ship owners could send their ships without bearing the risk for themselves and investors could diversify their risk by investing in multiple different ships and voyages.

The East India companies eventually began paying dividends from the proceeds collected from multiple voyages instead of creating single-time LLCs for each voyage. It was the first form of joint-stock companies in which the companies in which the companies could demand more capital, build larger fleets and provide larger returns for investors.

1.1.2 Equity Market:

An equity market is a hub in which shares of companies are issued and traded. The market comes in the form of an exchange- which facilitates the trade between buyers and sellers – or over-the-counter (OTC) in which buyers and sellers find each other.

It is important to begin with a good understanding of what is equity market in the Indian context. Equity markets, often called as stock market or share market, are a place where shares of companies or entities are traded. The market allows sellers and buyers to deal in equity or shares in the same platform.

In the global context, equities are traded either over the counter or at stock exchanges. There are multiple buyers and sellers of the same equity/share. Hence, you stand a good chance strike a nice deal at the equity market. If you want to being online equity trading in India, you have to get a demat account. Open a demat account in simple steps.

1.1.3 Equity Market in India:

Equities are mostly traded on the stock exchanges in India. In the Indian stock market, equities are available for trading at the National Stock Exchange (NSE), the Bombay Stock Exchange (BSE) and the latest entrant, Metropolitan Stock Exchange of India (MSE). Shares of stock market listed companies are bought/sold.

Equity share trading is roughly in two forms- spot/cash market and futures market. These are the different types of equity market in India. The spot market or cash market is a public financial market in which stocks are traded for immediate delivery. The futures market is a place where the share's delivery is due at a later date.

1.1.4 Equity Markets Work:

The concept behind how the stock market works is simple. Think of an auction house where buyers and sellers negotiate prices and make trades. Now, substitute the auction house and items with equity market and shares. Companies list their shares on an exchange. Investors can buy shares in the primary market i.e. IPOs, and secondary market.

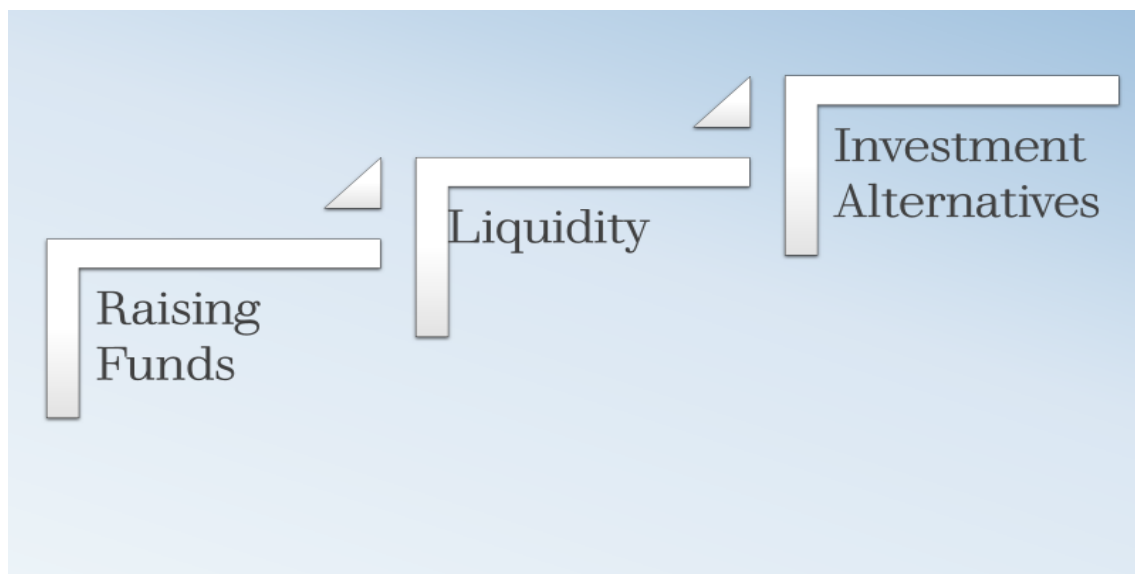
The stock market is regulated by a financial watchdog. The equity market is maintained by stock exchanges, and various stakeholders like brokers, dealers, clearing corporations etc. It is an extended family of institutions and this is the true equity market meaning.

1.1.5 Types of Equity Markets:

- **Primary Equity Market:** These are the shares offered to general investors through IPOs. Once the IPO is closed, the shares of a company are listed on the stock exchange. The two major stock exchanges facilitating trading in stocks are the NSE and BSE.
- **Secondary Equity Market:** If you do not purchase stocks of a company at the time of an IPO, you can purchase and sell the shares in the secondary market. Here, you plan your investment by deciding on an entry and exit point.

1.1.6 Significant of Equity Market:

In a market-based economy, equity markets are critical. They offer opportunities to obtain and increase funds, liquidity, and investment size.



These vital functions ensure that our economy continues to thrive and are the hallmark of capitalism.

1. Raising Funds:

It makes it easier to raise equity funds. This is critical for entrepreneurs who have a business idea but warrant the necessary finances to get it off the ground.

Banks are debt investors, and they are reluctant to advance loans to these companies without collateral or disproportionately high-interest rates.

As a result, giving up a stake in their company in exchange for backing is a financially sound idea for these entrepreneurs.

These companies may access the largest pools of cash through equity markets because they don't have to go out and find individual investors; rather, investors are brought to them through a network of investment banks and fiscal exchanges.

2. Liquidity

Market liquidity is also provided through it. The ease with which an asset can be converted into cash is referred to as liquidity. For example, checking accounts are the most liquid, but a painting is illiquid as it will typically take longer to sell.

Because equity markets are a centralized center for buyers and sellers, it is simple to identify someone eager to buy or sell your stock, and you can easily convert it into cash.

Because they reflect the immediate underlying supply and demand from millions of buyers and sellers worldwide, these markets are powerful price engines.

Stock prices rise when demand is high and buying activity is high, whereas stock prices fall when demand is low, and selling activity is high.

3. Investment Alternatives

For investors, equity markets offer a wide range of investment alternatives. By selecting different equity securities, investors can tailor their risk profile and gain exposure to various firms and industries.

These are also viable options for debt investing, which is advantageous to investors with larger risk tolerance.

These markets serve as a hub in which shares of companies are issued and traded. The market comes in the form of an exchange – which facilitates the trade between buyers and sellers – or over-the-counter (OTC), in which buyers and sellers find each other.

1.1.7 Trading in an Equity Market:

In the equity market, investors bid for stocks by offering a certain price, and sellers ask for a specific price. When these two prices match, a sale occurs. Often, there are many investors bidding on the same stock. When this occurs, the first investor to place the bid is the first to get the stock. When a buyer will pay any price for the stock, they are buying at market value; similarly, when a seller will take any price for the stock, they are selling at market value.

When a company offers its stock on the market, it means the company is publicly traded and each stock represents a piece of ownership. This appeal to investors, and when a company does well, its investors are rewarded as the value of their stocks rise.

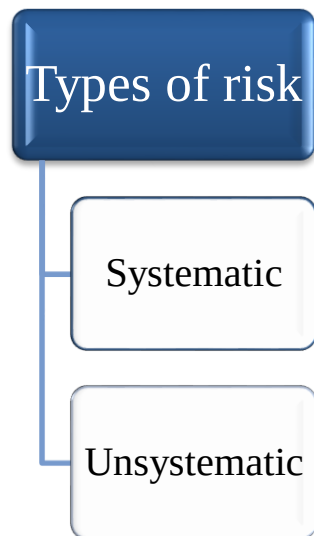
The risk comes when a company is not doing well, and its stock value may fall. Stocks can be bought and sold easily and quickly, and the activity surrounding a certain stock impacts its value. For example, when there is a high demand to invest in the company, the price of the stock tends to rise, and when many investors want to sell their stocks, the value goes down.

1.2 Introduction and Overview of Risk and Return:

The concept of **risk and return** makes reference to the possible economic loss or gain from investing in securities. A gain made by an investor is referred to as a return on their investment. Conversely, the risk signifies the chance or odds that the investor is going to lose money. In the case that an investor chooses to invest in an asset with minimal risk, the possible return then is often modest. In contrast, an investment with a high-risk component has a higher probability of generating larger profits.

1.2.1 Risks:

Whenever you invest or save, there are different types of risks that can be involved. But there are typically two categories that the risks are placed into: systematic risks and unsystematic risks.



1.2.1.1 Systematic:

Risks that can influence a complete economic market or at minimum a significant portion of it are known as **systematic risks**. They are the dangers of losing assets as a result of various macroeconomic or political risks which impact the general market performance. There are many types of systematic risks; a few of those are:

- **Political risk** - Political risk arises largely as a result of political insecurity in a nation or area. For example, if a country goes to war, the firms that operate there are deemed unsafe, and therefore risky.
- **Market risk** - Market risk is the byproduct of investors' overall inclination to follow the market. So it is essentially the inclination of security values to shift together.

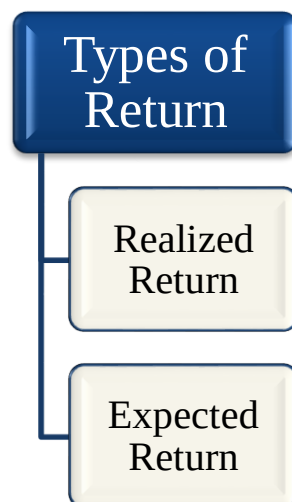
- **Exchange rate risk** - This type of risk arises from the unpredictability of currency value fluctuations. As a result, it impacts enterprises that conduct foreign exchange operations, such as export and import firms, or firms that do business in a foreign country.
- **Interest rate risk** - A shift in the market's rate of interest causes this type of risk. It mostly affects fixed-income assets since bond costs are connected to interest rates, but it also affects the valuation of stocks.

1.2.1.2 Unsystematic:

Unsystematic risk is a type of risk that impacts only one sector or one business. It is the danger of losing money on an investment because of a business or sector-specific hazard. A shift in leadership. A safety recall on a good, a legislative reform that might reduce firm sales or a new rival in the market is all examples of unsystematic risk.

1.2.2 Return:

There are two types of return that are most focused on: realized return and expected return.



1.2.2.1 Realized Return:

Realized return refers to the actual return on an investment over a specific time frame. It is critical to recognize that nothing can alter a realized return. It's really a post-fact number that no

action can alter. It merely provides information to investors to help them make wiser financial choices in the future.

1.2.2.2 Expected Return:

An expected return is the estimate of profits or losses that an investor may expect from an investment. The expected return is a metric used to estimate if an investment will have a positive or negative net outcome on average. The expected return is often founded on previous data and so cannot be guaranteed in the foreseeable future; yet, it frequently establishes acceptable expectations.

1.3 History of Financial Investment:

In the real world, individuals are neither completely rational nor irrational, they are normal persons having different kinds of temperament, attitude and behaviour and their response to a particular event is an outcome of these variables on which every individual is dependent upon. Research on financial investment has a long history. For decades, one of the most popular ideas in traditional finance literature is the Modern Financial Theory (MFT) which focuses on rational behaviour of investor and states that rationality exists in the behaviour of investors in financial markets with logical outcomes.

1.3.1 Investment:

An investment is an asset or item acquired with the goal of generating income or appreciation. Appreciation refers to an increase in the value of an asset over time. When an individual purchases a good as an investment, the intent is not to consume the good but rather to use it in the future to create wealth.

An investment always concerns the outlay of some resource today—time, effort, money, or an asset—in hopes of a greater payoff in the future than what was originally put in.

For example, an investor may purchase a monetary asset now with the idea that the asset will provide income in the future or will later be sold at a higher price for a profit.

1.4 Behavioural finance:

The field of behavioural finance attempts to combine finance and psychology in order to provide a plausible explanation for how financial markets work and how financial market agents behave. The emergence of this field can be attributed to the inability of traditional finance models to explain the empirical financial market patterns.

Behavioural finance proposes that the anomalies observed in the real world can be explained by relaxing some of the assumptions of traditional finance theories and models.

According to Statman (2012), in traditional finance, investors are rational whereas in behavioural finance, investors are normal. The field of behavioural finance takes a departure from the homo-economics assumption and acknowledges the fact that human beings are influenced by their emotions, cognitive limitations and many psychological factors. This is a more appropriate approach to provide a plausible explanation for how financial markets and its participants behave.

Investors are influenced by a variety of behavioural biases. Behavioural biases are rules of thumb or mental shortcuts that people use to simplify the decision-making process. These biases may lead to investment mistakes and wealth erosion. Behavioural biases can be divided into two categories; frame dependent biases and heuristic driven biases. Understanding these biases is very important in order to understand the psychology of investors. It is essential to understand investor psychology as psychology makes them susceptible to making biased decisions. These biased decisions can prove to be disastrous for the investors. If these mistakes occur on a wider level, it can lead to financial market crises such as the subprime crisis. In order to avoid such mistakes, it is imperative to investigate and identify which biases influence investors and how these biases may cause them to make mistakes. If investors are made aware about this phenomenon, they can avoid making such mistakes in the future. If investors are aware about the factors which drive their decision making, they can make better decisions in the future. Research about investor behaviour becomes imperative in this regard.

Behavioural finance has benefited us in realizing that the concept of homo economics is erroneous, thanks to the work of David Ricardo. Investor may be unable to maintain self-control, may be overconfident in their talents, may miscalculate facts, may overreact, or may simply follow the herd. The term regret refers to a feeling of sorrow or disappointment over something done or not done, several studies in the field of behavioural finance in India have Discovered the presence of these biases in Indian investors. As a result, Indian investors are not immune to the effects of their emotions and other psychological processes.

1.4.1 Behavioural Finance Key Themes:

Four key themes can be considered the foundations of behavioural finance. These themes are heuristics, framing, emotions, and market inefficiency.

1.4.1.1 Heuristics:

Heuristics are called rules of thumb that are used by people to reduce the cognitive efforts and resources required to find solutions to a complex problem. Heuristics are basically mental shortcuts that simplify the complex decision-making process. When Investors have to make decisions, they are exposed to many alternatives which are uncertain. Investors usually have limited ability to quantify the likelihood of the results associated with each of these alternatives. Under such a situation, investors tend to use rules of thumb or heuristics to simplify the decision-making process. Some of the popular heuristics are representativeness, anchoring, overconfidence, status quo, mental accounting etc. Behavioural finance believes that financial market practitioners make mistakes because they rely on heuristics to process data and to make financial decisions. One example of a popular rule of thumb is: 'past performance is the best predictor of future performance, so invest in a stock or mutual fund having the best performance in the last five years'.

1.4.1.2Framing:

Framing is another important theme in behavioural finance. Sometimes people make different choices for the same decision problem when the decision problem is framed differently. This

behaviour is referred to as frame dependence by the psychologists. Investors' predictions about the stock market differ depending on whether they are asked to forecast future prices or future returns (Glaser, Langer, Reyniers, & Weber, 2007). Behavioural Finance also states that investors' perceptions about risk and return are also greatly influenced by how decision problems are framed.

1.4.1.3 Emotions:

Human beings are emotional organisms. Human behaviour is influenced by emotions and sentiments. Behavioural finance acknowledges this fact. Moreover, the field endorses the role that Keynes' 'animal spirits' play in explaining investor choices and behaviour and how this shapes financial markets (Akerlof & Shiller, 2009). Our feelings determine our investment decisions and judgments in a subtle way. Emotional decision making may help in explaining why bubbles are created and why markets break down from time to time.

1.4.1.4 Market Inefficiency:

Traditional finance assumes market efficiency which means that security prices are determined by rational investors and reflect all the available information in the market. Therefore security prices are right and investors cannot beat the market. But research has shown that cognitive errors and behavioural biases of individual investors and groups of people influence the market prices and push them away from their fundamental value.

1.4.2 Applications of Behavioural Finance:

Behavioural finance doesn't focus only on finding and understanding the behaviour of financial market participants and how it affects markets. Behavioural finance also focuses on the possibility of overcoming these cognitive errors and helping investors to make better decisions. Awareness about behavioural biases can help people to avoid their investment mistakes. Following points highlight the applications of behavioural finance.

1.4.2.1 Markets:

Markets are a mechanism for distributing wealth in the society. The health and future prospects of an economy are directly affected by the functioning of markets. The field of behavioural finance studies how behavioural biases and cognitive mistakes of market participants affect the functioning of market. Behavioural finance studies why markets might be inefficient and how to make markets more efficient.

1.4.2.2 Corporations:

In corporations, managers or a group of top level officers make crucial decisions involving a lot of money. If these managers or top level officers are prone to behavioural biases, it will affect shareholders and many other stake holders in the corporation. The biases of top level executives cannot be corrected by arbitrage. This makes behavioural finance even more important for corporate finance. Managers have to take many important decisions about capital budgeting, dividend policy, financing, corporate governance, mergers and acquisitions etc.

1.4.2.3 Regulatory Environment:

Behavioural finance has applications in regulatory environment and policymaking as well. The heuristics and behavioural biases that affect investors, managers and market can also affect policy and law makers.

1.4.2.4. Individual Investors:

A lot of research has been done to document the biases of individual investors and how these biases affect the financial wealth of individual investors. Researchers have also attempted to identify factors which cause individual investors to take biased decisions. Factors such as cognitive limitations, experience and learning have been found to play an important role in financial decision-making.

1.5 Behavioural bias:

Behavioural biases are systematic, predictable errors or influences that apply to everyone when they interpret information and make decisions. Understanding behavioural biases means you

can understand how consumers make predictable mistakes when choosing and using financial products. This means that some clients may be choosing products that do not meet their needs or provide the best price or value.

1.5.1 Behavior biases are important for Investors:

When an investor decides to invest in something, he or she will look for the highest possible reward while minimizing risk. Investing, on the other hand, is not a matter of theory and analysis. Prior to making an investment, an investor must examine a variety of significant internal and external elements. While the relevance of external factors is clear, experts have begun to recognize the importance of internal factors in investing in the last decade. This category encompasses behavioural and psychological variables. As a result of the ensuing biases and tendencies, investors behave irrationally, which has an effect on their decision-making behavior.

1.5.2 TPES BEHAVIOR BIASES OF INVESTORS:

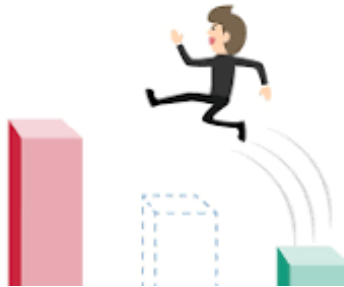
1.5.2.1 Loss aversion bias:



When compared to other sorts of clients, conservative investors are more sensitive to the agony of losses than the joy of wins. Even when there is no hope of a return on investment, some

clients may hold on to their losses for too long. Many financial advisors have a tendency to be fearful about losing money.

1.5.2.2 Overconfidence bias:



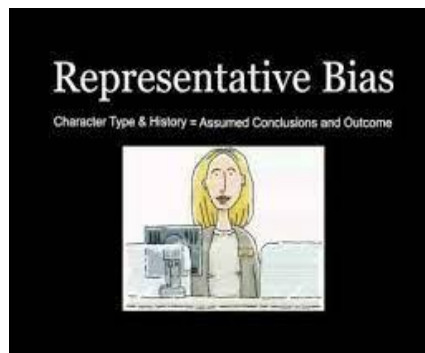
It's better to think of overconfidence as having an unjustified faith in one's own abilities and thinking, which includes both mental and emotional components. Investors who are overconfident in the quality of their judgments show their overconfidence in their own abilities. Overconfident individuals also tend to believe that the information they possess is superior. Because of their belief of having superior skills and superior information, overconfident investors are willing to take more risk.

1.5.2.3 Regret Aversion Bias:



If you're a moderate investor, you're more likely to put off making a decision for fear that you'll regret it later. It is possible for moderate investors to be too cautious in their investment decisions due to past losses.

1.5.2.4 Representative bias:



When new information is processed, a faulty perceptual framework can lead to a prejudice known as representativeness bias. Some investors try to predict outcomes that are similar to their own preconceived notions in order to make the new information easier to comprehend. Representative bias is seen in investor behaviour when investors judge a company based on the recent performance of that company's stock or when investors consider companies which are perceived to have competent managers or good recent returns to be good companies and invest in those companies.

1.5.2.5 Anchoring bias:



Asked if investor should buy or sell an investment, conservative investors are more likely to rely on purchase points or arbitrary price levels. Anchoring may cause financial market participants to take incorrect decisions. Anchoring may lead investors to take decisions based on irrelevant facts and statistics.

1.5.2.6 Confirmation bias:



Confirmation bias is an investor's tendency to believe and seek information that supports his perceived notion and avoid others. Because of confirmation bias, the investor may neglect information which does not match his belief.

1.5.2.7 Herding behavior:



It is refers to the tendency of investors or traders to follow the actions of their peers, rather than making independent decisions based on their own analysis and information. Herding is a very powerful bias. When investors start mimicking each others 'actions, the Cumulative impact of their herd behaviour is intensified and it may lead to stock market bubbles or crashes depending on the direction of the price movement.

CHAPTER:-2

Literature Review

This chapter provides a review of literature on individual investor behaviour and behavioural biases. First of all research on behavioural biases has been discussed. Various studies on factors affecting individual investor behaviour have also been explored. At the end, the research gaps and objectives of the study have been highlighted.

Studies on Behavioural Biases of Individual Equity Investors:

In this section studies related to behavioural biases of individual equity investors are divided into two parts- studies conducted in abroad and studies conducted in India.

1Hirshleifer (2001) described the role of behavioural biases (narrow framing, mental accounting, representativeness heuristic and overconfidence) in asset pricing in securities market by analyzing the available literature. The study concluded that behavioural biases lead to imperfect rationality in stock market.

2. Strong (2003) analyzed the home bias among equity fund managers from the United States, continental Europe, the United Kingdom, Japan, the Asia- Pacific basin, and South Africa. Results of the study indicated that fund managers from the United States, the United Kingdom, continental Europe and Japan show a significant relative optimism towards their home equity market, thus they have home bias.

3. Jamshidinavid ET. al. (2012) analyzed the relationship between Personality traits and demographics on the financial behaviour prejudices (overconfidence, herding and disposition effect) in Tehran Stock Market in Iran using Structure Equation Modeling (SEM). Findings exposed that age has inverse relationship with herding behaviour. It means that whenever age of investor becomes older, his/her herd behavior becomes lower. Women show more herding than men. Overconfidence has positive relationship with education. It means that higher education is increasing the level of confidence in investing. Residential area has no relationship with behavioural biases (overconfidence, herding and disposition effect).

4.Vijaya (2014) identified five major factors (Overconfidence, Anchoring, Loss Aversion, Herd behaviour and Market factors.) that can influences retail investor's investment behaviour in Indian stock market. Data was collected from 200 individual investors of Hyderabad and Secunderabad. Factor analysis, Mean and Standard Deviation were used to analyse the data.

Analysis showed 34 that Anchoring, Loss Aversion, Herd behaviour and Market factors have moderate impact of investment decisions while overconfidence has high impact on investment decisions of individual equity investors in Indian stock market.

5. Rajagopalan and Gurusamy (2015) analyzed the relationship between personality traits and behavioural biases of retail investors in Chennai (India). Factor analysis by principal component method has been applied to reduce the number of psychological biases and personality traits into ten meaningful factors and seven meaningful factors respectively. The correlation analysis was used to find out the relationship between personality traits and psychological biases. Personality factors were labeled as gregariousness, Self-consciousness, Risk-aversion, Diligence, Pragmatism, Aesthetic and Altruism. Psychological biases factors were labeled as Self-enhancement bias, Cognitive dissonance, Ambiguity aversion, Illusion of control, Extrapolation bias, Performance attribution bias, Competency bias, Information Overload bias, SocioConformity bias and Disposition Effect. Findings revealed that three personality types (Diligence, Self-conscious investors and Pragmatic investors) are susceptible to all psychological biases.

6. Prosad ET. al. (2015) found the existence of behavioural biases (Overconfidence, Excessive optimism (pessimism), Herd behaviour and Disposition effect) among Indian investors. Data was collected from 500 individual investors of Delhi-NCR. Findings concluded that all studied behavioural biases are present among Indian investors and on ranking these biases in their order of prevalence, overconfidence comes out to be the most important bias in the Indian equity market.

7. Kumar and Goyal (2016) investigated the relationship between rational decision-making (Demand identification, Information search and Evaluating alternatives) and behavioural biases (Overconfidence bias, Disposition effect and Herding bias) among individual investors in India. Data was collected from 386 individual investors of Jaipur. Statistical techniques like t-test, analysis of variance (ANOVA) and Fisher's least significant difference (LSD) test have been used in this study.

8. .Mushinada and Veluri (2018) analyzed that the relationship between investors rationality and behavioral biases (self-attribution, overconfidence). Structural equation modeling (SEM)

was used to understand the individual investors are being rational to self-attribution bias and overconfidence bias. The findings revealed that there is a significant positive covariance between self-attribution and overconfidence it indicated that an increase/decrease in self-attribution results in the increase/decrease in overconfidence. It is also witnessed that the personal characteristics of an individual investor such as gender, age, occupation, annual income and their trading experience have an impact on behavioural biases.

9. Chandra and Kumar (2011) found that individual investor behaviour is motivated by a variety of psychological heuristics and biases. They document that the behaviour of Indian individual investors is affected by anchoring, representativeness, mental accounting and overconfidence. Prosad (2014) found that Indian investors are affected by optimism, overconfidence, disposition effect and herd behaviour.

10. Dickason and Ferreira (2018) established a relationship between risk tolerance level, investor's personality and behavioural biases of investors in South Africa. Descriptive survey method was used to obtain the objectives of the study. A questionnaire was developed to know the risk tolerance level, investor's personality and behavioural biases of investors. Nine behavioural finance biases (Representativeness, Overconfidence, Anchoring, Gamblers fallacy, Availability bias, Loss aversion, Regret aversion, Mental accounting and self-control) were transformed into statements and respondents were asked to rank the statements from most to Least relevant. The findings of the study revealed that medium-risk tolerant investors were categorized as moderate-to-growth investor's subject towards anchoring, regret aversion, representativeness, overconfidence, gamblers fallacy and the availability bias. Investors within this category will make investment decisions based on previous incorrect financial decisions. Investors with a low-risk tolerance level and categorized as conservative investors were subject towards the loss aversion and mental accounting bias. Investors subject towards the self-control bias were the only investors to fall in the aggressive personality category and have a high-risk tolerant level. Findings revealed that all the behavioural factors exist among individual investors of South Africa while making investment decisions. Results showed that overconfidence, anchoring, prospect variables (loss aversion and regret aversion) and herding variables have moderate impact while prospect variable (mental accounting) and market variables have high impact on investment decisions.

CHAPTER:-3

Research Methodology

3.1 Problem statement:

“To study the existence of behavioural biases among individual equity investors, examine the differences in behavioural biases of individual equity investors on the basis of demographic profile and study the impact of behavioural biases on the investment decisions of individual equity investors.”

3.2 Objectives of the study:

- To study the impact of demographic profile on various factors affecting the behavioural biases of investment decision.
- To study behavioural biases affecting investment decisions in equity market.
- To analyse the impact of behavioural biases on the investment decisions of individual equity investors.
- To identify which behavioural bias is most prominent among individual investors of the Indian equity Market.

3.3 Research Design:

Research design of the study is descriptive in nature. A descriptive research approach is justified because it aims to identify and explain the factors present in a given scenario and describe the relationship between these variables to provide a picture of the situation.

3.4 Data collection:

3.4.1 Type of data:

- **Primary sources:**

In research used primary sources to get data for this study. It is original information gathered by the respondents. Primary data provide more confidence because it was obtained for a specific purpose.

3.4.2 Data collection method:

- **Questionnaire Method**

.There is a method for collecting primary data **Method of the questionnaire** for this study.

3.5 Target Population:

The objective of the study is to analyze the impact of behavioural biases on individual investor behaviour in Surat. Hence the target population for the study would be all the individual equity investors living in Surat city.

3.6 Sampling Unit:

The sampling unit for the study is the individual equity investor who is a resident of Surat and who makes equity share buy-sell-hold decisions on his/her own.

3.7 Sample size:

Sample size' is a market research term used for defining the number of individuals included in conducting research. Researchers choose their sample based on demographics, such as age, gender questions or physical location. The sample size of the study is **150 equity investors**.

3.8 Sampling techniques:

Convenience sampling technique is used to collect data from the respondent of this study.

3.9 Data analysis:

3.9.1 Tools and techniques are used:

IBM SPSS (trial version 29) and **MS EXCEL**, there are mention statistical software are used for the data management and analysis for the purpose of hypothesis testing and fulfillment of the objective.

Statistical methods used in this study are Descriptive statistics-Mean, S.D., One sample's' test, Independent sample't' test, Analysis of Variance (ANOVA).

Limitation of the study:

This study only concerning equity investor in Surat city. And only 150 individual equity investors' respondent concerning this study.

Due to time constraints, this study concentrates only on the loss aversion, overconfidence, regret aversion, representation, anchoring, confirmation, hear behaviour biases out of thirty biases in investor.

CHAPTER:-4

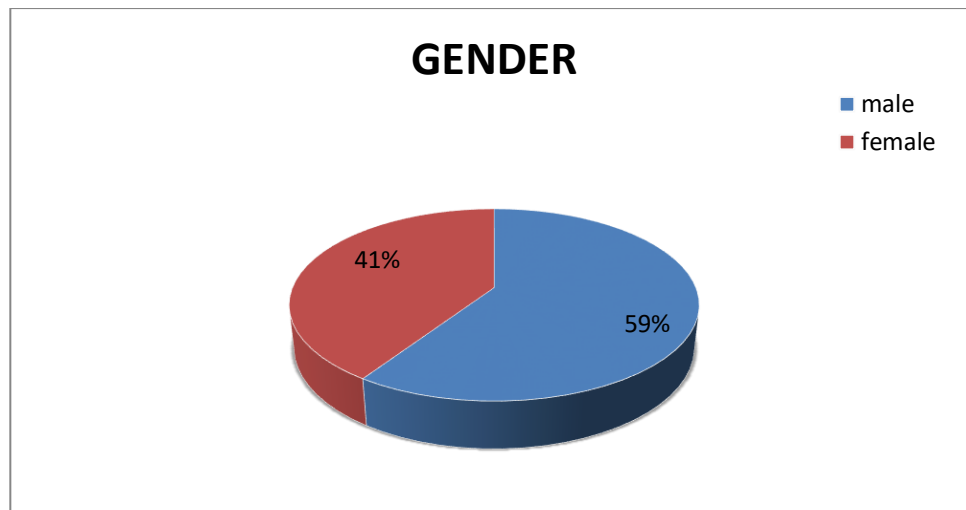
Data analysis

4.1 Graphical analysis of demographic factors:

4.1.1 GENDER CLASSIFICATION:

gender

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	male	89	59.3	59.3	59.3
	female	61	40.7	40.7	100.0
	Total	150	100.0	100.0	



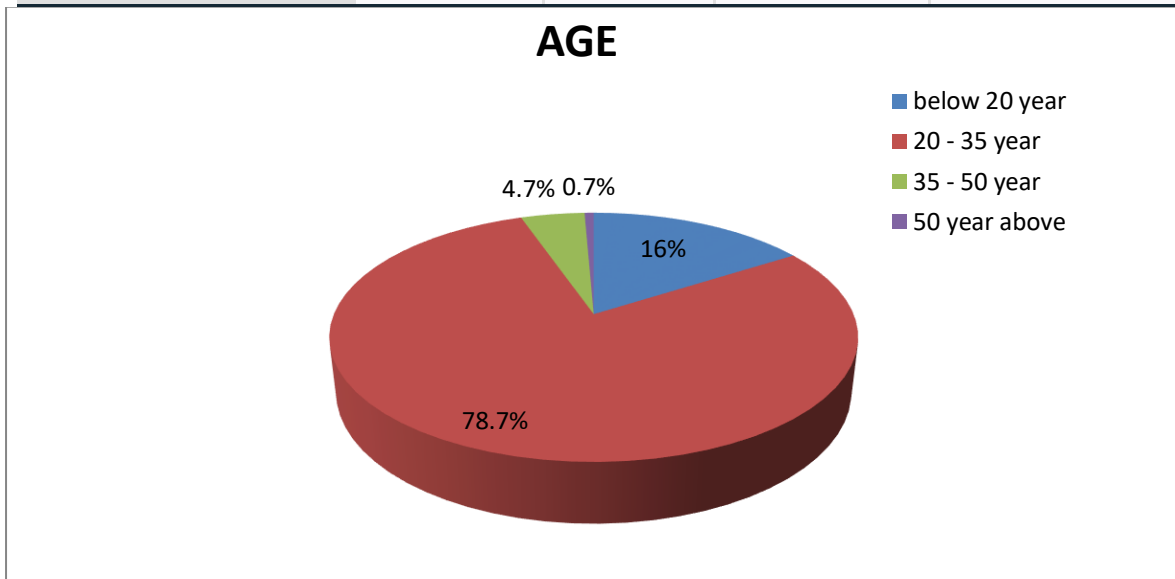
Interpretation:

The demographic profile of respondent suggests that the respondent on a gender basis there are 41% of the female respondent and approximately 59% male respondents. Most of the respondents of gender are male concerning 89 respondents.

4.1.2 AGE CLASSIFICATION:

age

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	below 20 year	23	15.3	15.3	15.3
	20-35 year	119	79.3	79.3	94.7
	35-50 year	7	4.7	4.7	99.3
	50 year above	1	.7	.7	100.0
	Total	150	100.0	100.0	



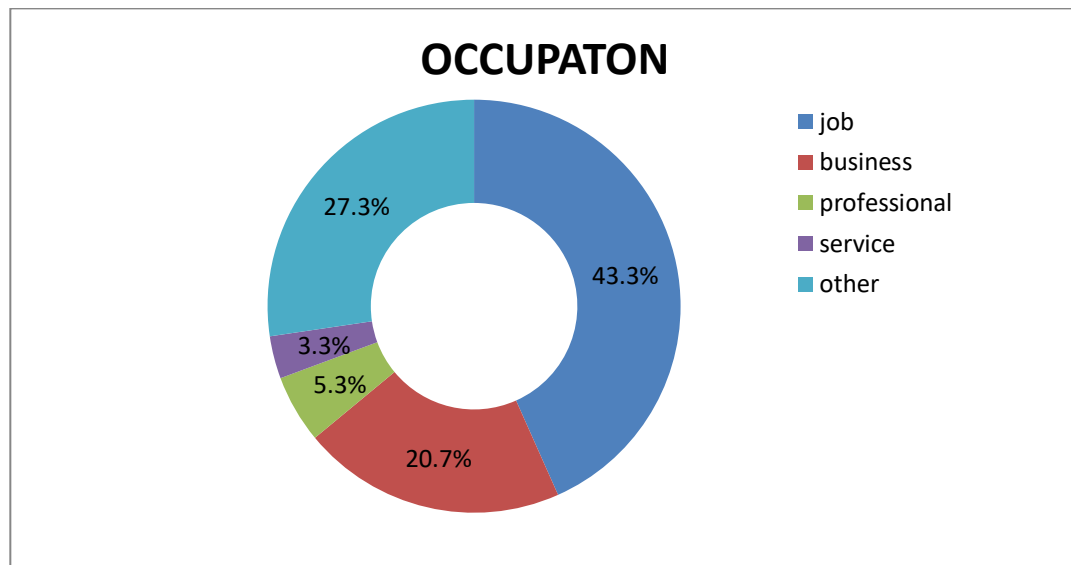
Interpretation:

Above graph is shows the, majority of investor's age is between 20-35 years that is 78.7% of investors are from 20-35 years out of 150 respondents, 16% of respondent's age is below 20 years, 4.7% of respondent's age is 35-50 years and 0.7% of respondent's age is above 50 years. That shows the majority of investors belong to age group of 20-35 years and they are more attractive to investing in equity market.

4.1.3 OCCUPATION CLASSIFICATION:

occupation

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	job	65	43.3	43.3	43.3
	business	31	20.7	20.7	64.0
	professional	8	5.3	5.3	69.3
	service	5	3.3	3.3	72.7
	other	41	27.3	27.3	100.0
	Total	150	100.0	100.0	



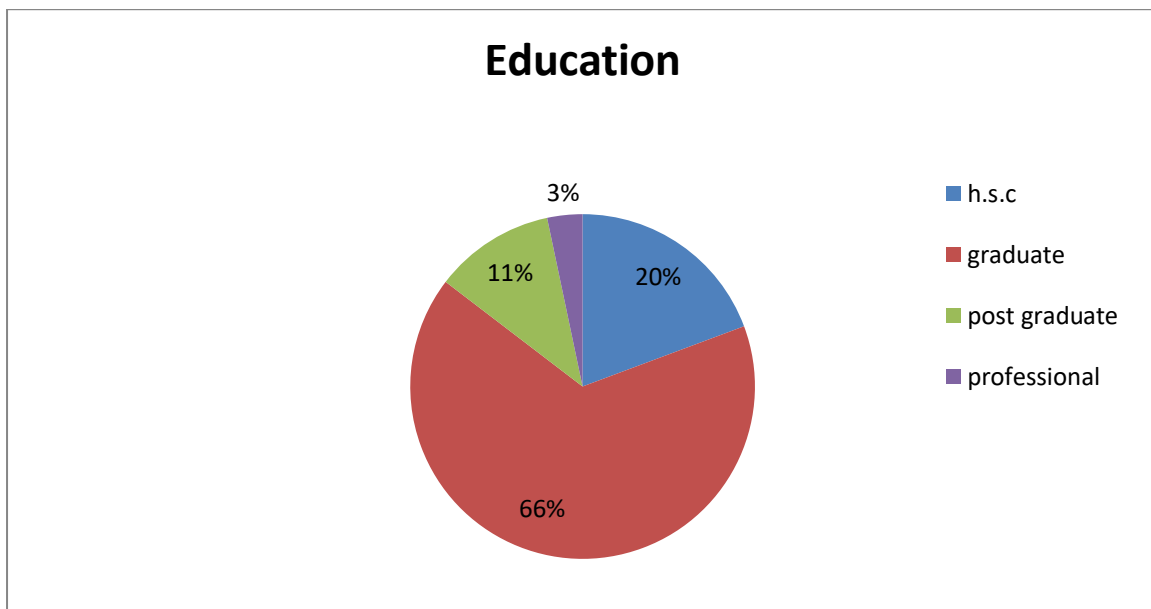
Interpretation:

The data reveals the distribution of occupations in a sample of 150 individual equity investors. The majority of individuals are engaged in jobs 43.3%, followed by those in business 20.7% and professional roles 5.3%. A smaller proportion is involved in service occupations 3.3%, and a diverse range of other occupations is represented by 27.3% of the sample. The data reflects occupational diversity within the sample, with jobs being the most prevalent category, followed by business and other varied occupations.

4.1.4 EDUCATION CLASSIFICATION:

education qualification

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	h.s.c	29	19.3	19.3	19.3
	graduate	99	66.0	66.0	85.3
	post graduate	17	11.3	11.3	96.7
	professional	5	3.3	3.3	100.0
	Total	150	100.0	100.0	



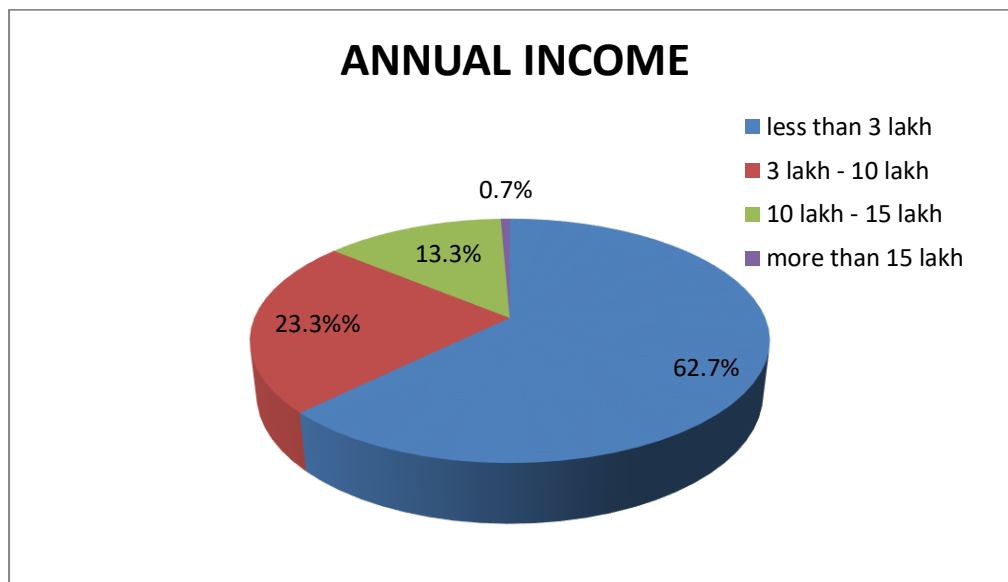
Interpretation:

Above table shows the classification of the basis of their educational qualification. The education of the respondents has significant impact on the investment behaviour to investing in equity market. Out of 150 respondents 66% are graduate, 11% are post graduate, 19.3% are school final and 3% are professional.

4.1.5 ANNUAL INCOME:

annual income

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	less than 300000	94	62.7	62.7	62.7
	300001 – 1000000	35	23.3	23.3	86.0
	1000001 – 1500000	20	13.3	13.3	99.3
	more than 1500001	1	.7	.7	100.0
	Total	150	100.0	100.0	



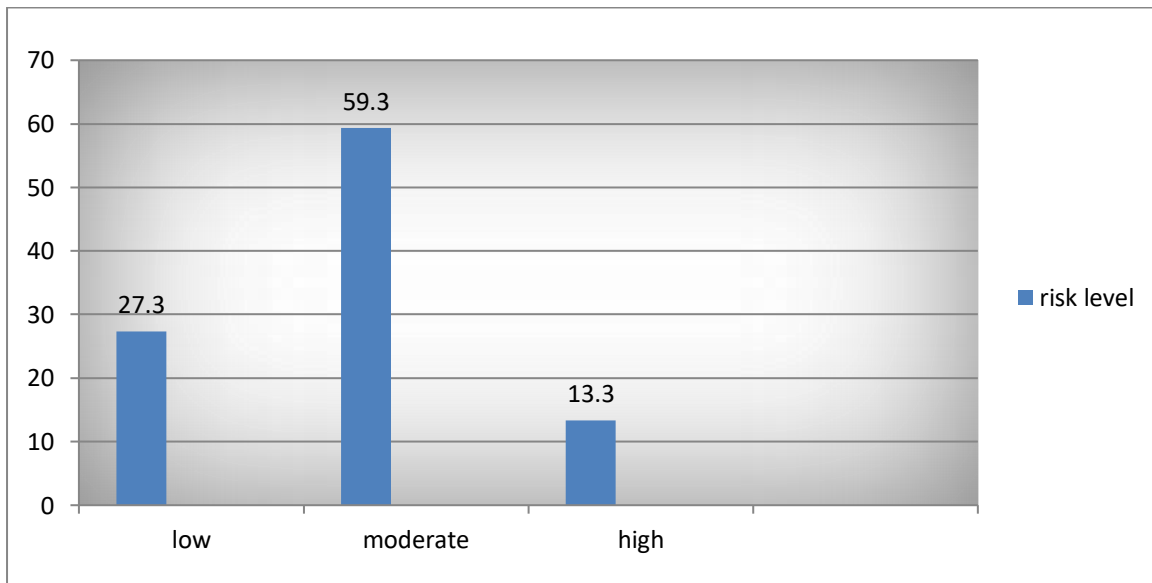
Interpretation:

Above information shows the annual income of equity investors. As above table in majority of investor's fall in the income category of less than 300000 and they are 62.7%. Out of 150 individual equity investors 23.3% of the respondents fall under the income category of 300001-1000000, 13.3% of the respondents fall under the 1000001-1500000 and 0.7% fall under the category of 1500001 above.

4.1.6 RATE THE RISK ASSOCIATED WITH INVESTMENT:

risk associated

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	low	41	27.3	27.3	27.3
	moderate	89	59.3	59.3	86.7
	high	20	13.3	13.3	100.0
	Total	150	100.0	100.0	

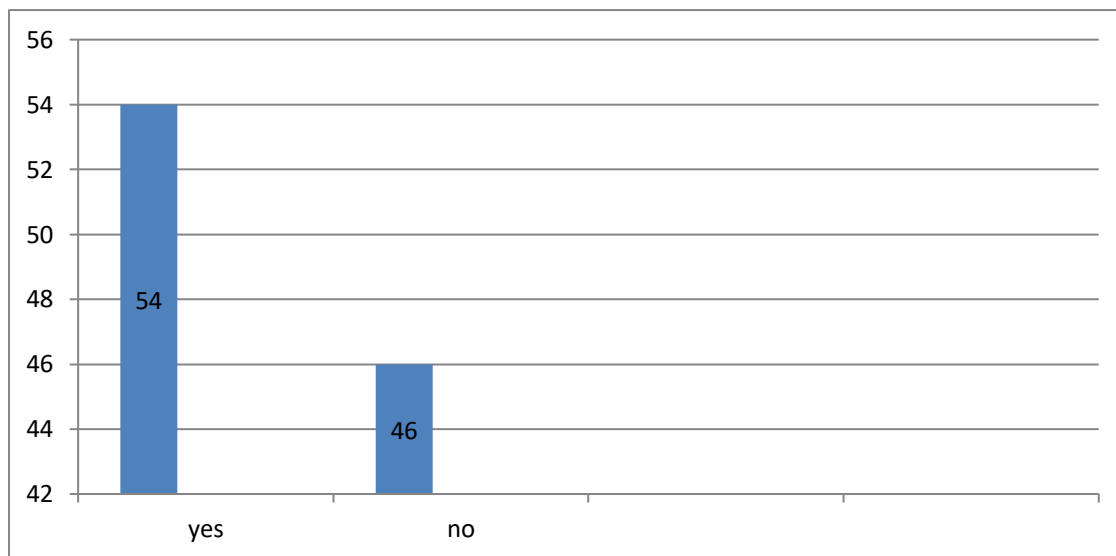


Interpretation:

As above the information most individual equity investors in the sample perceive a moderate level of risk, while a notable portion sees the risk as low, and a smaller group views it as high. The cumulative percentages indicate that 86.7% of the sample falls within the low and moderate risk categories.

4.1.7 Investment experience in stock market:

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	no	69	46.0	46.0	46.0
	yes	81	54.0	54.0	100.0
	Total	150	100.0	100.0	



Interpretation:

Above data shows the 69 respondents, which accounts for 46% of the total sample, reported having no investment experience in the stock market. 81 respondents, which accounts for 54% of the total sample, reported having investment experience in the stock market. This data suggests that a slight majority of respondents (54%) have investment experience in the stock market, while the remaining respondents (46%) do not have such experience.

4.2 Descriptive statistics of behavioural biases:

	BIASES	statement	MEAN	SD
overconfidence	OC1	I have sufficient knowledge of India stock market.	2.87	1.42
	OC2	I feel that I can on average, predict future share prices better than others.	2.97	1.16
	OC3	I take the responsibility of managing my portfolio and I trust my decision.	3.26	1.30
	OC4	I make all the important share investment decisions myself.	3.15	1.32
representativeness	RE5	I believe that I can forecast the future value of the stock on the basis of its past performance.	3.09	1.28
	RE6	I try to avoid investing in companies with a history of poor earning	3.08	1.25
	RE7	I use trend analysis of some to make investment decision.	3.25	1.35
	RE8	One of my primary indicators of investing in a company is its level of social responsibility.	3.08	1.21
Anchoring bias	AN9	I will choose the target price for buying / selling before commencing a transaction.	3.13	1.28
	AN10	When I decide to sell a stock, I keep its purchase price in mind.	3.37	1.34
	AN11	I am unlikely to buy a stock that was more expensive than last year.	2.89	1.21
	AN12	In a falling market I hold a losing stock till its price returns to its purchase level.	3.05	1.20
Confirmation bias	CB13	Before, buying a share, I ignore the information in the market that conflict with mine.	2.50	1.31
	CB14	Before, buying a share I appreciate the information in the market that support mine.	3.23	1.31

Heard behaviour	HB15	My selection of stock type depends on others investment.	2.91	1.22
	HB16	I prefer to buy/sell stock that witnessed many buying/selling orders during the trading day.	3.08	1.13
	HB17	I would invest in a stock by following my friend's recommendations.	2.78	1.21
	HB18	When you don't have information of a stock, you usually follow what majority of investors are doing.	2.86	1.28
Regret aversion	RA19	I avoid investing in profitable assets if I have incurred losses in similar investment in past.	2.83	1.15
	RA20	I often feel regret for selling a winning stock too early.	3.11	1.18
	RA21	I avoid selling shares if their value comes down.	3.02	1.25
	RA22	I became risk-seeking because I have made profits in the past.	3.05	1.22
Loss aversion	LA23	I use tool to avoid losses and protect the investments.	3.24	1.31
	LA24	I have many fears regarding the losses in buying and selling stocks.	2.95	1.22
	LA25	I never sell an investment at a loss with an expectation that it will eventually improve.	3.11	1.30
	LA26	I prefer low risk investment over risky investment even if their return in low.	3.03	1.29

The table shows the descriptive statistics i.e. mean and S.D of each statement of behavioural biases. It is evident from the table that mean value of all the statements measuring **behavioural biases is more than 3, which shows that individual equity investors are biased during trading in stock market.**

4.2.1 Hypothesis testing:

Hypothesis 1: Behavioural biases exist among individual equity investors

H1a: Overconfidence bias exists among individual equity investors

H1b: Representation bias exists among individual equity investors

H1c: Anchoring bias exists among individual equity investors

H1d: Confirmation bias exist among individual equity investors

H1e: Heard behaviour bias exist among individual equity investors

H1f: Regret Aversion bias exist among individual equity investors

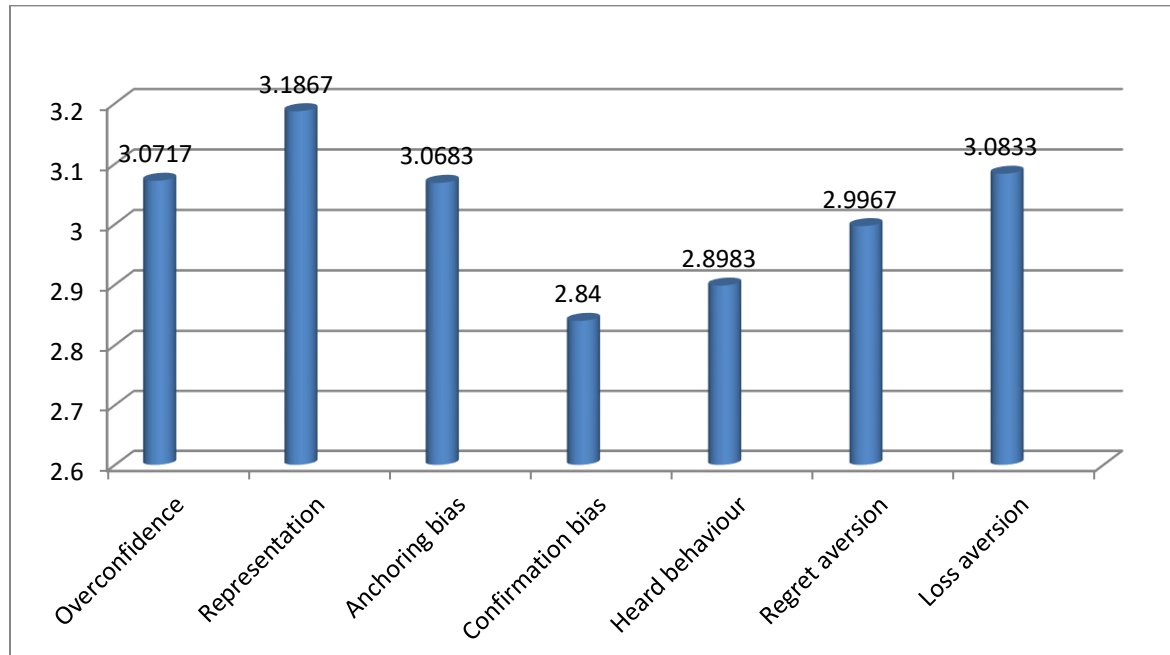
H1g: Loss Aversion bias exist among individual equity investors

One sample t test (test value=2.5)

One-Sample Statistics				
Biases	N	Mean	Std. Deviation	Std. Error Mean
overconfidence	150	3.0717	1.12564	.09191
representation	150	3.1867	1.18972	.09714
Anchoring bias	150	3.0683	1.03824	.08477
Confirmation bias	150	2.8400	1.05124	.08583
Heard behaviour	150	2.8983	.96416	.07872
Regret aversion	150	2.9967	.99495	.08124

Loss aversion	150	3.0833	1.02103	.08337
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One-Sample Statistics							
	Test Value = 2.5						
	t	df	Significance		Mean Difference	95% Confidence Interval of the Difference	
			One-Sided p	Two-Sided p		Lower	Upper
Overconfidence	6.220	149	<.001	<.001	.57167	.3901	.7533
Representation	7.069	149	<.001	<.001	.68667	.4947	.8786
Anchoring bias	6.704	149	<.001	<.001	.56833	.4008	.7358
Confirmation bias	3.961	149	<.001	<.001	.34000	.1704	.5096
Heard behaviour	5.060	149	<.001	<.001	.39833	.2428	.5539
Regret aversion	6.114	149	<.001	<.001	.49667	.3361	.6572
Loss aversion	6.997	149	<.001	<.001	.58333	.4186	.7481



Mean of behavioural biases

The value of t at 5% level of significance for one tailed test is 1.645 and at 1% level of significance for one tailed test is 2.358. 't' value for all behavioural biases is more than the critical value of t (2.358) at 1% level of significance for one tailed test. So, hypotheses H1a, H1b, H1c, H1d, H1e, H1f, and H1g are accepted and it can be concluded that all the studied behavioural biases exist among individual equity investors.

4.3 Analysis of differences in behavioural biases of individual equity investors on the basis of demographic profiles:

This objective examines the differences in behavioural biases of individual equity investors on the basis of demographic profile. The objective is fulfilled with help of independent sample t test and one way ANOVA (one way analysis of variance).following hypotheses are framed to achieve the objective.

4.3.1 Independent sample t test

4.3.1.1 Gender and Behavioural biases

Hypothesis 2: There exists significant difference between behavioural biases among individual equity investors on the basis of gender

Based on the above hypothesis following sub-hypotheses are framed-

H2a: overconfidence bias differ among individual equity investors on the basis of gender

H2b: representation bias differ among individual equity investors on the basis of gender

H2c: Anchoring bias differ among individual equity investors on the basis of gender

H2d: Confirmation bias differ among individual equity investors on the basis of gender

H2e: Heard behaviour bias differ among individual equity investors on the basis of gender

H2f: Regret aversion bias differ among individual equity investors on the basis of gender

H2g: Loss aversion bias differ among individual equity investors on the basis of gender

Independent Sample "t" Test to test the Difference in Behavioural Biases on the Basis of Gender							
	gender	N	Mean	Std. Deviation	Std. Error Mean	T Statistic (p value)	significance / not significance
overconfidence	male	89	3.1348	1.15432	.12236	0.829 (0.491)	Not significance
	female	61	2.9795	1.08522	.13895		
representation	male	89	3.2416	1.09197	.11575	0.628 (0.884)	Not significance
	female	61	3.1066	1.32481	.16962		
Anchoring bias	male	89	3.1545	1.07628	.11409		

A study on impact of individual equity investor's behavioural biases to investment decision on equity market in surat region

	female	61	2.9426	.97510	.12485	1.230 (0.035)	significance
Confirmation bias	male	89	2.8483	1.11397	.11808	0.117 (0.037)	Significance
	female	61	2.8279	.96127	.12308		
Heard behaviour	male	89	3.0253	.98428	.10433	1.966 (0.253)	Not significance
	female	61	2.7131	.91040	.11656		
Regret aversion	male	89	3.0618	1.08795	.11532	0.968 (0.020)	Significance
	female	61	2.9016	.84073	.10764		
Loss aversion	male	89	3.1292	1.12254	.11899	0.664 (0.007)	significance
	female	61	3.0164	.85619	.10962		

Interpretation:

The provided data compares various cognitive biases between male and female groups, presenting the mean, standard deviation, standard error mean, t-statistic, and p-value for each bias.

While there are no significant differences in overconfidence, representation and heard behavior between males and females because their p value is more than 0.05, so H2a, H2b and H2e are rejected. There are significant differences in anchoring bias, Confirmation bias, regret aversion and loss aversion bias between males and females because their p value is less than 0.005, so H2c, H2d, H2f, and H2g are accepted. Males show slightly higher levels of anchoring bias and regret aversion, while females demonstrate slightly higher levels of loss aversion.

4.3.1.2 Marital status and Behavioural biases:

Hypothesis 3: There exists significant difference between behavioural biases among individual equity investors on the basis of marital status.

Based on the above hypothesis following sub-hypotheses are framed-

H3a: overconfidence bias differs among individual equity investors on the basis of marital status.

H3b: representation bias differs among individual equity investors on the basis of marital status.

H3c: Anchoring bias differs among individual equity investors on the basis of marital status.

H3d: Confirmation bias differs among individual equity investors on the basis of marital status.

H3e: Heard behaviour bias differ among individual equity investors on the basis of marital status.

H3f: Regret aversion bias differs among individual equity investors on the basis of marital status.

H3g: Loss aversion bias differs among individual equity investors on the basis of marital status.

Independent Sample “t” test to test the Difference in Behavioural Biases on the basis of Marital Status							
Bias name	Marital status	N	Mean	Std. Deviation	Std. Error Mean	T Statistic (p value)	Significance/ not significance
overconfidence	married	44	3.1932	1.18966	.17935	0.851 (0.054)	Significance
	unmarried	106	3.0212	1.09985	.10683		
representation	married	44	3.3011	1.10966	.16729	0.758 (0.053)	Not significance
	unmarried	106	3.1392	1.22332	.11882		
Anchoring bias	married	44	3.0852	1.04679	.15781		

	unmarried	106	3.0613	1.03958	.10097	0.128 (0.639)	Not significance
Confirmation bias	married	44	2.8977	1.03755	.15642	0.432 (0.091)	Not significance
	unmarried	106	2.8160	1.06083	.10304		
Heard behaviour	married	44	2.9205	1.03116	.15545	0.180 (0.488)	Not significance
	unmarried	106	2.8892	.93991	.09129		
Regret aversion	married	44	3.1307	1.09104	.16448	1.063 (0.079)	Significance
	unmarried	106	2.9410	.95216	.09248		
Loss aversion	married	44	3.2045	1.06650	.16078	0.936 (0.044)	significance
	unmarried	106	3.0330	1.00242	.09736		

Significant at 5% level of significance:

Interpretation:

The provided data presents the mean, standard deviation, standard error, t-statistic, and p-value for various cognitive biases across different marital statuses.

While there are no significant differences in representation, anchoring, confirmation and heard behavior between married and unmarried individuals because their p value is more than 0.05, so H3b, H3c, H3d and H3e are rejected. There are significant differences in overconfidence, regret aversion and loss aversion bias between males and females because their p value is less than 0.005, so H3a, H3f and H3g are accepted.

There are some marginal differences observed in certain cognitive biases between married and unmarried individuals, most of the differences are not statistically significant. However, married individuals tend to exhibit slightly higher levels of overconfidence, representation, confirmation bias, regret aversion, and loss aversion compared to unmarried individuals.

4.3.2 One way analysis of variance (ANOVA):

4.3.2.1 Age and behavioural biases

Hypothesis 4: There exists significant difference between behavioural biases among individual equity investors on the basis of their age.

Based on the above hypothesis following sub-hypotheses are framed-

H4a: overconfidence bias differs among individual equity investors on the basis of their age.

H4b: representation bias differs among individual equity investors on the basis of their age.

H4c: Anchoring bias differs among individual equity investors on the basis of their age.

H4d: Confirmation bias differs among individual equity investors on the basis of their age.

H4e: Heard behaviour bias differ among individual equity investors on the basis of their age.

H4f: Regret aversion bias differs among individual equity investors on the basis of their age.

H4g: Loss aversion bias differs among individual equity investors on the basis of their age.

One Way Analysis of Variance (ANOVA) to test the Difference in Behavioural Biases and Age Group

Biases	Age	N	Mean	Std. Deviation	F statistic (p value)	Significance/ not significance
Overconfidence	below 20 year	23	2.7391	1.15915	1.369 (0.022)	Significance
	21-35 year	119	3.1324	1.13207		
	36-50 year	7	3.3214	.68791		
	51 year and above	1	1.7500	.		

A study on impact of individual equity investor's behavioural biases to investment decision on equity market in surat region

	Total	150	3.0717	1.12564		
representation	below 20 year	23	3.3587	1.82599	0.445 (0.721)	Not significant
	21-35 year	119	3.1513	1.06972		
	36-50 year	7	3.3571	.31810		
	51 year and above	1	2.2500	.		
	Total	150	3.1867	1.18972		
Anchoring bias	below 20 year	23	2.9783	1.05774	0.233 (0.873)	Not significance
	21-35 year	119	3.0945	1.05715		
	36-50 year	7	2.8571	.74801		
	51 year and above	1	3.5000	.		
	Total	150	3.0683	1.03824		
Confirmation bias	below 20 year	23	2.6304	1.02489	0.856 (0.046)	Significance
	21-35 year	119	2.8487	1.06077		
	36-50 year	7	3.2857	.99403		
	51 year and above	1	3.5000	.		
	Total	150	2.8400	1.05124		
Heard behaviour	below 20 year	23	2.9022	.85193	1.116 (0.041)	significance
	21-35 year	119	2.9097	1.00303		
	36-50 year	7	2.7500	.76376		
	51 year and above	1	2.5000	.		

	Total	150	2.8983	.96416		
Regret aversion	below 20 year	23	2.8587	.85222	1.005 (0.039)	significance
	21-35 year	119	2.9874	1.03018		
	36-50 year	7	3.4643	.74202		
	51 year and above	1	4.0000	.		
	Total	150	2.9967	.99495		
Loss aversion	below 20 year	23	3.0217	.93818	0.056 (0.982)	Not significance
	21-35 year	119	3.1008	1.05381		
	36-50 year	7	3.0000	.88976		
	51 year and above	1	3.0000	.		
	Total	150	3.0833	1.02103		

Significant at 5% level of significance:

Interpretation:

The provided data presents the means, standard deviations, standard errors, and (ANOVA) F-statistics for different age groups in relation to various cognitive biases. Here's the interpretation for each cognitive bias within different age groups.

Significant differences are found in overconfidence, confirmation bias, heard behavior, and regret aversion across different age groups because their p value < 0.05 , so H4a, H4d, H4e and H4f are accepted. While representation, anchoring bias, and loss aversion do not show significant variations because their p value > 0.05 so, H4b, H4c and H4g are rejected. These results provide insights into how cognitive biases may vary based on individuals' age groups.

4.3.2.2 Educational qualification and behavioural biases

Hypothesis 5: There exists significant difference between behavioural biases among individual equity investors on the basis of their educational qualification.

Based on the above hypothesis following sub-hypotheses are framed-

H5a: overconfidence bias differs among individual equity investors on the basis of their educational qualification.

H5b: representation bias differs among individual equity investors on the basis of their educational qualification.

H5c: Anchoring bias differs among individual equity investors on the basis of their educational qualification.

H5d: Confirmation bias differs among individual equity investors on the basis of their educational qualification.

H5e: Heard behaviour bias differ among individual equity investors on the basis of their educational qualification.

H5f: Regret aversion bias differs among individual equity investors on the basis of their educational qualification.

H5g: Loss aversion bias differs among individual equity investors on the basis of their educational qualification.

One Way Analysis of Variance (ANOVA) to test the Difference in Behavioural Biases and educational qualification

Biases	education	N	Mean	Std. Deviation	F Statistic (P value)	Significance / not significance
overconfidence	h.s.c	29	2.4741	1.17895	5.636 (.001)	Significance
	graduate	99	3.1035	1.04735		
	post graduate	17	3.7059	1.09771		
	professional	5	3.7500	.93541		
	Total	150	3.0717	1.12564		

A study on impact of individual equity investor's behavioural biases to investment decision on equity market in surat region

representation	h.s.c	29	2.6897	1.20204	2.709 (0.047)	Significance
	graduate	99	3.2424	1.17394		
	post graduate	17	3.6176	1.17632		
	professional	5	3.5000	.72887		
	Total	150	3.1867	1.18972		
Anchoring bias	h.s.c	29	2.7845	1.27252	1.429 (0.237)	Not Significance
	graduate	99	3.0859	.95442		
	post graduate	17	3.4265	1.08168		
	professional	5	3.1500	.82158		
	Total	150	3.0683	1.03824		
Confirmation bias	h.s.c	29	2.6379	1.22399	1.337 (0.265)	Not Significance
	graduate	99	2.8182	.98316		
	post graduate	17	3.1471	1.16946		
	professional	5	3.4000	.65192		
	Total	150	2.8400	1.05124		
Heard behaviour	h.s.c	29	2.6034	1.10502	1.750 (0.159)	Not Significance
	graduate	99	2.9470	.89662		
	post graduate	17	3.2059	1.06520		
	professional	5	2.6000	.80234		
	Total	150	2.8983	.96416		
Regret aversion	h.s.c	29	2.5603	1.09311	3.394 (0.020)	Significance
	graduate	99	3.0303	.90863		
	post graduate	17	3.4412	1.17769		
	professional	5	3.3500	.57554		
	Total	150	2.9967	.99495		
Loss aversion	h.s.c	29	2.6897	1.11134	2.340 (0.049)	Significance
	graduate	99	3.1288	.95789		
	post graduate	17	3.4559	1.13961		
	professional	5	3.2000	.85513		
	Total	150	3.0833	1.02103		

Significant at 5% level of significance:

Interpretation:

The provided data displays the means, standard deviations, standard errors, and (ANOVA) F-statistics for different levels of education qualifications in relation to various cognitive biases.

Significant differences are found in overconfidence, representation, heard behavior, and regret aversion across different education qualifications because their p value is less than 0.05 so; H5a, H5b, H5e, H5f and H5g are accepted.

While anchoring bias, confirmation bias, and loss aversion do not show significant variations because their p value is more than 0.05 so, H5c, H5d, and H5ge are rejected. These results provide insights into how cognitive biases may vary based on individuals' education levels.

There are significant differences in overconfidence, representation, and regret aversion across different groups, while no significant differences are observed for anchoring bias, confirmation bias, heard behavior, and loss aversion.

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4.3.2.3 Income and behavioural biases

Hypothesis 6: There exists significant difference between behavioural biases among individual equity investors on the basis of their annual income.

Based on the above hypothesis following sub-hypotheses are framed-

H6a: overconfidence bias differs among individual equity investors on the basis of their annual income

H6b: representation bias differs among individual equity investors on the basis of their annual income

H6c: Anchoring bias differs among individual equity investors on the basis of their annual income

H6d: Confirmation bias differs among individual equity investors on the basis of their annual income

H6e: Heard behaviour bias differ among individual equity investors on the basis of their annual income

H6f: Regret aversion bias differs among individual equity investors on the basis of their annual income

H6g: Loss aversion bias differs among individual equity investors on the basis of their annual income

One Way Analysis of Variance (ANOVA) to test the Difference in Behavioural Biases and annual income

Biases	Income	N	Mean	Std. Deviation	F statistic (p value)	Significance /not significance
overconfidence	less than 30000	94	2.8989	1.12302	2.392 (0.051)	significance
	300001 - 1000000	35	3.3000	1.01605		
	1000001 - 1500000	20	3.5125	1.20436		
	more than 1500001	1	2.5000	.		
	Total	150	3.0717	1.12564		
representation	less than 300000	94	2.9920	1.06915	2.642 (0.042)	Significance
	300001 - 1000000	35	3.6143	1.48193		
	1000001 – 1500000	20	3.3750	1.00164		
	more than 1500001	1	2.7500	.		
	Total	150	3.1867	1.18972		
Anchoring bias	less than 300000	94	2.9947	1.08940	0.484 (0.694)	Not significance
	300001 - 1000000	35	3.2214	.87399		
	100001 - 1500000	20	3.1250	1.09274		
	more than 1500001	1	3.5000	.		
	Total	150	3.0683	1.03824		

A study on impact of individual equity investor's behavioural biases to investment decision on equity market in surat region

Confirmation bias	less than 300000	94	2.7553	1.07199	0.989 (0.400)	Not significance
	300001 - 1000000	35	2.9286	.97877		
	100001 - 1500000	20	3.1250	1.07453		
	more than 1500001	1	2.0000	.		
	Total	150	2.8400	1.05124		
Heard behaviour	less than 300000	94	2.8138	.97408	0.891 (0.448)	Not significance
	300001 - 1000000	35	3.0786	.96018		
	1000001 – 1500000	20	3.0125	.93004		
	more than 1500001	1	2.2500	.		
	Total	150	2.8983	.96416		
Regret aversion	less than 300000	94	2.8564	.97653	2.418 (0.049)	Significance
	300001 – 1000000	35	3.1214	.91228		
	1000001 - 1500000	20	3.4625	1.11295		
	more than 1500001	1	2.5000	.		
	Total	150	2.9967	.99495		
Loss aversion	less than 300000	94	2.9787	1.00513	1.218 (0.030)	significance
	300001 - 1000000	35	3.1786	.95788		
	1000001 - 1500000	20	3.4250	1.17848		
	more than 1500001	1	2.7500	.		
	Total	150	3.0833	1.02103		

Significant at 5% level of significance:

Interpretation:

The provided data presents the means, standard deviations, standard errors, and (ANOVA) F-statistics for different income brackets in relation to various cognitive biases.

Significant differences are found in representation, regret aversion, and loss aversion across different income brackets because their p value is less than 0.05 so; H6a, H6b, H6f and H6g are accepted. While overconfidence, anchoring bias, confirmation bias, and heard behavior do not show significant variations based on income brackets because their p value is more than 0.05 so; H6b, H6c and H6e are rejected. These results provide insights into how cognitive biases may vary based on individuals' income levels.

4.3.2.4 Occupation and behavioural biases

Hypothesis 7: There exists significant difference between behavioural biases among individual equity investors on the basis of their occupation

Based on the above hypothesis following sub-hypotheses are framed-

H7a: overconfidence bias differs among individual equity investors on the basis of their occupation

H7b: representation bias differs among individual equity investors on the basis of their occupation

H7c: Anchoring bias differs among individual equity investors on the basis of their occupation

H7d: Confirmation bias differs among individual equity investors on the basis of their occupation

H7e: Heard behaviour bias differ among individual equity investors on the basis of their occupation

H7f: Regret aversion bias differs among individual equity investors on the basis of their occupation

H7g: Loss aversion bias differs among individual equity investors on the basis of their occupation

One Way Analysis of Variance (ANOVA) to test the Difference in Behavioural Biases and occupation

Biases	Occupation	N	Mean	Std. Deviation	F statistic (p value)	Significance/ not significance
overconfidence	job	65	3.0038	1.15793	1.062 (0.142)	Not significance
	business	31	3.1129	1.15981		
	professional	8	3.8438	.98141		
	service	5	3.1000	.96177		
	other	41	2.9939	1.08179		
	Total	150	3.0717	1.12564		
representation	job	65	3.1654	1.41026	1.236 (0.049)	significance
	business	31	3.3629	1.06231		
	professional	8	3.7813	.68709		
	service	5	3.5000	.58630		
	other	41	2.9329	.97632		
	Total	150	3.1867	1.18972		
Anchoring bias	job	65	2.9308	1.06846	1.462 (0.031)	significance
	business	31	3.2339	1.06641		
	professional	8	3.7813	.82848		
	service	5	3.0500	.54199		
	other	41	3.0244	1.01520		
	Total	150	3.0683	1.03824		
Confirmation	job	65	2.9000	1.03908		

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bias	business	31	3.0806	1.23218	1.303 (0.022)	significance
	professional	8	3.0000	.46291		
	service	5	2.6000	1.08397		
	other	41	2.5610	.97593		
	Total	150	2.8400	1.05124		
Heard behaviour	job	65	2.8808	1.00692	0.527 (0.716)	Not significance
	business	31	3.0403	1.02692		
	professional	8	3.0938	1.07685		
	service	5	3.1000	.67546		
	other	41	2.7561	.86691		
	Total	150	2.8983	.96416		
Regret aversion	job	65	2.9077	.96478	1.422 (0.022)	significance
	business	31	3.3306	1.11502		
	professional	8	3.2188	.77272		
	service	5	3.1000	.87678		
	other	41	2.8293	.97217		
	Total	150	2.9967	.99495		
Loss aversion	job	65	3.0385	1.05163	0.785 (0.253)	Not significance
	business	31	3.2258	1.16253		
	professional	8	3.5625	.47716		
	service	5	2.8500	.65192		
	other	41	2.9817	.96726		
	Total	150	3.0833	1.02103		

Significant at 5% level of significance:

Interpretation:

The provided data presents the means and statistical significance for various cognitive biases among different occupational groups.

Significant differences are found in representation, anchoring bias, confirmation bias, and regret aversion across different occupational groups because their p-value is less than 0.05 so, H7b, H7c, H7d and H7f are accepted. However, no significant differences are observed in overconfidence, heard behavior, and loss aversion based on occupation of individuals because their p-value more than 0.05 so, H7a, H7e and H7g are rejected. These results suggest that certain cognitive biases may vary based on the occupation of individuals, providing insights into how occupational factors may influence decision-making tendencies.

4.3.2.5 Rate the risk associate with investment and behavioural biases:

Hypothesis 8: There exists significant difference between behavioural biases among individual equity investors on the basis of their risk associated capacity.

Based on the above hypothesis following sub-hypotheses are framed-

H8a: overconfidence bias differs among individual equity investors on the basis of their risk associated capacity.

H8b: representation bias differs among individual equity investors on the basis of their risk associated capacity.

H8c: Anchoring bias differs among individual equity investors on the basis of their risk associated capacity.

H8d: Confirmation bias differs among individual equity investors on the basis of their risk associated capacity.

H8e: Heard behaviour bias differ among individual equity investors on the basis of their risk associated capacity.

H8f: Regret aversion bias differs among individual equity investors on the basis of their risk associated capacity.

H8g: Loss aversion bias differs among individual equity investors on the basis of their risk associated capacity.

One Way Analysis of Variance (ANOVA) to test the Difference in Behavioural Biases and risk associate with investment

Biases	Risk rate	N	Mean	Std. Deviation	F Statistic (p value)	Significance \ Not significance
overconfidence	low	41	2.3110	1.11773	21.775 (<.001)	significance
	moderate	89	3.2107	.95785		
	high	20	4.0125	.88658		
	Total	150	3.0717	1.12564		
representation	low	41	2.5854	1.07472	12.169 (<.001)	Significance
	moderate	89	3.2725	1.16509		
	high	20	4.0375	.89691		
	Total	150	3.1867	1.18972		
Anchoring bias	low	41	2.5061	1.06285	11.458 (<.001)	significance
	moderate	89	3.1910	.94224		
	high	20	3.6750	.90721		
	Total	150	3.0683	1.03824		
Confirmation bias	low	41	2.3780	1.07110	10.857 (<.001)	Significance
	moderate	89	2.8764	.96907		
	high	20	3.6250	.88667		
	Total	150	2.8400	1.05124		
Heard behaviour	low	41	2.4756	.93174		
	moderate	89	2.9607	.89815		

	high	20	3.4875	.97830	8.670 (<.001)	significance
	Total	150	2.8983	.96416		
Regret aversion	low	41	2.4756	1.01212	14.018 (<.001)	Significance
	moderate	89	3.0618	.85137		
	high	20	3.7750	1.00296		
	Total	150	2.9967	.99495		
Loss aversion	low	41	2.5305	1.00498	17.266 (<.001)	significance
	moderate	89	3.1320	.88956		
	high	20	4.0000	.91047		
	Total	150	3.0833	1.02103		

Significant at 5% level of significance:

Interpretation:

The provided data presents the means, standard deviations, standard errors, and (ANOVA) F-statistics for different levels of perceived risk (low, moderate, and high) in relation to various cognitive biases.

For each behavioural bias, there are significant differences in mean scores across different levels of perceived risk. Their p-value less than 0.05 so, H8a, H8b, H8c, H8d, H8e, H8f and H8g are accepted. This suggests that individuals' cognitive biases vary depending on their perception of risk, indicating an intricate relationship between risk perception and decision-making tendencies.

CHAPTER:-5

Finding

Finding:

In this section major findings of the study are discussed.

Frequency analysis of investor's demographic factors:

DEMOGRAPHIC CLASSIFICATION		frequency	percentage(%)
GENDER	male	89	59.3
	female	61	40.7
AGE	below 20 year	23	15.3
	21-35 year	119	79.3
	36-50 year	7	4.7
	51 year above	1	0.7
OCCUPATION	job	65	43.3
	business	31	20.7
	professional	8	5.3
	service	5	3.3
	other	41	27.3
EDUCATION	h.s.c	29	19.3
	graduate	99	66
	post graduate	17	11.3
	professional	5	3.3
ANNUAL INCOME(rs)	less than 300000	94	62.7
	300001-1000000	35	23.3
	1000001-1500000	20	13.3
	more than 1500001	1	0.7
RISK CAPABILITIES	low	41	27.3
	moderate	89	59.3
	high	20	13.3
INVEATMENT EXPERIENCE	no	69	46
	yes	81	54

Descriptive statistics of behaviour biases:

Descriptive statistics: MEAN = <3 behavioural biases is more than 3, which shows that individual equity investors are biased during trading in stock market.

HYPOTHESIS TESTING:

One sample t test: (behaviour biases)

't' value for all behavioural biases is more than the critical value of t (2.358) at 1% level of significance for one tailed test. It can be concluded that all the studied behavioural biases exist among individual equity investors.

Independent sample t test:

- **Gender and behaviour biases:** Males show slightly levels of anchoring bias and regret aversion, while females demonstrate slightly higher levels of loss aversion.
- **Marital status and behavioural biases:** Most of the differences are not statistically significant. However, married individuals tend to exhibit slightly higher levels of overconfidence, representation, confirmation bias, regret aversion, and loss aversion compared to unmarried individuals.

ANOVA:

- **Age and behavioural biases:** Significant differences are found in overconfidence, confirmation bias, heard behavior, and regret aversion across different age groups, there are accepted. While representation, anchoring bias, and loss aversion do not show significant variations with age group there are rejected. These results provide insights into how cognitive biases may vary based on individuals' age groups.
- **Educational qualification and behavioural biases:** There are significant differences in overconfidence, representation, and regret aversion across different groups, while no significant differences are observed for anchoring bias, confirmation bias, heard behavior, and loss aversion.
- **Income and behavioural biases:** Significant differences are found in representation, regret aversion, and loss aversion across different income brackets. While overconfidence, anchoring bias, confirmation bias, and heard behavior do not show significant variations based on income brackets

- **Occupation and behavioural biases:** representation, anchoring bias, confirmation bias, and regret aversion across different occupational groups are accepted. No significant differences are observed in overconfidence, heard behavior, and loss aversion based on occupation of individuals
- **Rate the risk associate with investment and behavioural biases:** suggests that individuals' cognitive biases vary depending on their perception of risk, indicating an intricate relationship between risk perception and decision-making tendencies.

Chapter:-6

Suggestion

SUGGESTION:

In India the behavioural biases are mostly studied using survey based techniques, leaving a huge opportunity for researchers to use secondary data. Therefore, efforts can be made to detect the possible market indicators that can predict the presence and impact of behavioural biases with the help of secondary data.

The present study have examined the behavioural biases of individual equity investors, similar study may be conducted on institutional investors, foreign institutional investors (FII) and high net worth investors who have tremendous influence over Indian stock markets.

There are over thirty biases in Behavioural Finance which have been studied in the rest of the world. In the future, researchers could look into other biases which are yet to be studied in India and get a more complete picture of the investors 'behaviour.

Behavioural biases of female investors may also be studied in order to provide a more comprehensive picture of the psychological profile of the female investors.

Chapter:-7

Conclusion

CONCLUSION:

Individual equity investors are affected by various behavioural biases regardless of demographic factors. These biases can significantly impair investment decision-making, highlighting the importance of awareness, education, and strategies to mitigate their impact for more rational and successful investing. Certain biases such as anchoring, representativeness, overconfidence, and herding tend to increase with the age of the investor. Moreover, the degree of biases tends to rise with higher levels of education. This suggests that older and more educated investors may need to be particularly cautious about these biases.

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ANNEXURE:

A study on impact of individual investor's behaviour biases to investment decision on equity market in Surat region

Dear Respondent

I am **VALA URMILA M** a student of **TY BBA Finance** from **S.V. Patel College of computer science and business management**.

I am doing research on “A study on impact of individual investor's behaviour biases to investment decision on equity market in surat city” I want valuable response regarding above topic, so can please spend few minutes from your busy schedule to answer the following questions. I would like to assure you that this study is only done for education purpose and the data provided by you will be kept confidential.

Kindly give your genuine and honest answer in this survey and help me complete my research work.

Demographics questions:

1. Name:

2. Gender: 1 Male
 2 Female
 3 Other

3. Age: 1. Below 20 year
 2. 21-35 year
 3. 36-50 year
 4. 51 year and above

4. Marital status: 1. Married
 2. Unmarried

5. Occupation:
1. Job
 2. Business
 3. Professional
 4. Service
 5. Other
6. Annual income:
1. less than 300000
 2. 300001 - 1000000
 3. 1000001 – 1500000
 4. More than 1500001
7. Qualification:
1. H.S.C
 2. Graduate
 3. Post graduate
 4. Professional
8. Do you have sufficient financial management knowledge?
1. Yes
 0. No
9. How do you rate the risk associated with investment?
1. Low
 2. Moderate
 3. High
10. Do you have Investment experience in stock market?
1. Yes
 0. No

Questions representing behavioural biases of an individual: Please evaluate the degree of your agreement with the impact of behavioural factors on your investment decision making:

1. Strongly disagree
2. Disagree
3. Neutral
4. Agree
5. Strongly disagree

NO.	STATEMENTS					
		1	2	3	4	5
OC 1.	I have sufficient knowledge of India stock market.					
2.	I feel that I can on average, predict future share prices better than others.					
3.	I take the responsibility of managing my portfolio and I trust my decision.					
4.	I make all the important share investment decisions myself.					
RP5 .	I believe that I can forecast the future value of the stock on the basis of its past performance.					
6.	I try to avoid investing in companies with a history of poor earning.					
7.	I use trend analysis of some to make investment decision.					
8.	One of my primary indicators of investing in a company is its level of social responsibility.					
AB 9.	I will choose the target price for buying / selling before commencing a transaction.					
10.	When I decide to sell a stock, I keep its purchase price in mind.					
11.	I am unlikely to buy a stock that was more expensive					

	than last year.					
12.	In a falling market I hold a losing stock till its price returns to its purchase level.					
CB 13.	Before, buying a share, I ignore the information in the market that conflict with mine.					
14.	Before, buying a share I appreciate the information in the market that support mine.					
HB 15.	My selection of stock type depends on others investment.					
16.	I prefer to buy/sell stock that witnessed many buying/selling orders during the trading day.					
17.	I would invest in a stock by following my friend's recommendations.					
18.	When you don't have information of a stock, you usually follow what majority of investors are doing.					
RA 19.	I avoid investing in profitable assets if I have incurred losses in similar investment in past.					
20.	I often feel regret for selling a winning stock too early.					
21.	I avoid selling shares if their value comes down.					
22.	I became risk-seeking because I have made profits in the past.					
LA 23.	I use tool to avoid losses and protect the investments.					
24.	I have many fears regarding the losses in buying and selling stocks.					
25.	I never sell an investment at a loss with an expectation that it will eventually improve.					
26.	I prefer low risk investment over risky investment even if their return is low.					